

# Week 1 Submission Multimodal AI Tutor Literature Survey and Corpus Selection

## Existing Models

### Academic Educational AI Tutors:

#### Khanmigo (Khan Academy):

A conversational learning assistant powered by GPT-4 fine-tuned with Socratic instructional prompting. It acts as a pedagogical guide rather than an answer generator. However, its retrieval architecture is grounded exclusively on course transcripts and textual exercises, omitting video visual tracks and slide diagrams.

#### Similarities to our project:

- **Pedagogical focus:** Both systems prioritize guiding the learner's understanding rather than acting as a standard search engine or answer generator.
- **Domain-specific grounding:** Both restrict the AI's knowledge base to approved instructional material to prevent open-ended hallucinations.

#### Main differences:

- **Unimodal vs. Multimodal:** Khanmigo relies exclusively on text (course transcripts and written exercises). Your project will process both the audio transcript *and* the visual frames, allowing it to answer questions about diagrams, code diffs, or whiteboard drawings visible on screen.
- **Memory Architecture:** Khanmigo uses standard conversational context windows. Your project introduces a strict policy-controlled learner memory that separates ephemeral chat state from durable mastery information, complete with user correction/deletion capabilities.
- **Evaluation:** Your blueprint requires a rigorous adversarial evaluation harness specifically testing for retrieval failure and prompt injection, which is not exposed or structurally guaranteed in Khanmigo's public architecture.

#### Coursera Coach:

An integrated course assistant that draws from lecture transcripts, reading assignments, and syllabus outlines. It provides coarse module-level references rather than sub-second video navigation and operates entirely on textual transcripts.

#### Similarities to our project:

- **RAG Foundation:** Both systems utilize Retrieval-Augmented Generation to fetch relevant course documents before generating an answer.
- **Boundary Enforcement:** Both systems are designed to stay strictly within the confines of the provided curriculum.

### Main differences:

- **Citation Granularity:** Coursera Coach provides coarse, high-level references (e.g., pointing a student to "Week 3"). Your project requires precise, sub-second video timestamp citations mapping directly to the exact moment a claim is supported.
- **Visual Blindness:** Like Khanmigo, Coursera Coach processes video strictly through text transcripts. It cannot "see" the slides the instructor is presenting.
- **Source-Version Invalidation:** Your project explicitly scopes for source-version invalidation (knowing when course material is outdated or replaced), whereas Coursera Coach treats the current syllabus as static text.

### Video Summarization:

#### Google NotebookLM:

A grounded document synthesis assistant that ingests PDFs, text files, and YouTube URLs. While it provides strict text passage citations, its YouTube ingestion processes only the closed-caption audio transcript; it does not process the video stream, rendering it blind to on-screen slides, code diffs, and whiteboard annotations.

#### Similarities to our project:

- **Strict Source Citations:** NotebookLM is the closest existing model to your citation requirement. It forces the LLM to show the exact evidence behind its claims.
- **Explicit Abstention:** NotebookLM excels at saying "I don't know" when the answer is not in the source material, aligning perfectly with your requirement to avoid confidently inventing unsupported answers.

### Main differences:

- **YouTube Ingestion Method:** When NotebookLM ingests a video, it only processes the closed-caption audio transcript. Your project requires true multimodal ingestion, linking the audio transcript to the actual visual frames to support queries that depend on visual evidence.
- **Learner State:** NotebookLM is a stateless research tool; it does not track learner mastery, durable profile information, or progress over time. Your project requires a distinct learner memory system.
- **Evaluation Suite:** Your project mandates a rigorous RAG evaluation harness (including tests for stale content and prompt injection) built directly into the system's architecture, rather than relying entirely on the underlying foundation model's general safety tuning.

# Literature Survey

## Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

**Methodology:** Lewis et al. (2020) established the foundational end-to-end RAG architecture combining pre-trained parametric memory (a sequence-to-sequence transformer, specifically BART) with explicit non-parametric memory accessed via a dense neural retriever (Dense Passage Retrieval; DPR). The paper introduced two core mathematical formulations: RAG-Sequence, which conditions generation across the entire output sequence on the same top-k retrieved documents, and RAG-Token, which marginalizes over different retrieved document latent variables at each generation step. Both formulations are trained end-to-end where retrieval probabilities and sequence generation log-likelihoods are jointly optimized without direct document supervision.

**Advantages & Disadvantages:** The major advantage is that externalizing factual knowledge enables dynamic updating and replacement of the non-parametric knowledge base without expensive retraining of the generator, significantly reducing hallucinations and yielding state-of-the-art results across open-domain QA (Natural Questions, WebQuestions, CuratedTREC). However, the primary disadvantage of this foundational framework is its exclusive reliance on single-vector dense bi-encoders, which frequently fail on exact keyword, code-syntax, or identifier queries where lexical matchers excel. In addition, marginalizing over latent documents per token incurs substantial inference latency, rendering real-time conversational streaming computationally demanding.

## VersionRAG: Version-Aware Retrieval-Augmented Generation for Evolving Documents

**Methodology:** VersionRAG (Huwiler et al., 2025) formalizes the problem of document evolution in technical knowledge bases, where standard RAG frameworks suffer acute performance degradation due to temporal insensitivity. The authors introduce a hierarchical graph structure that models document histories by capturing version sequences, explicit content boundaries, and granular change deltas across document states. During the retrieval phase, VersionRAG routes user prompts through specialized paths driven by intent classification, allowing the engine to execute version-aware filtering, diff inspection, and multi-version comparative tracking rather than relying purely on unconstrained semantic vector similarity.

**Advantages & Disadvantages:** The primary advantage of VersionRAG is its dramatic gain in precision on time-sensitive queries; on the curated VersionQA benchmark, it achieves 90% accuracy compared to 58% for naive RAG and 64% for GraphRAG, while solving implicit change detection at 60% accuracy where baselines virtually collapse (0–10%). Furthermore, its structural diffing eliminates full-graph re-indexing, consuming 97% fewer tokens during index ingestion than GraphRAG. The core disadvantage lies in its reliance on structured, trackable version boundaries and intent routing accuracy; for unstructured multimedia courses or loosely organized video lecture releases where change manifests across visual slide updates rather than discrete text diffs, constructing the version graph introduces parsing overhead and potential routing bottlenecks.

## Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods

**Methodology:** Cormack et al. (2009) proposed Reciprocal Rank Fusion (RRF), an unsupervised rank aggregation algorithm designed to combine document rankings from disparate information retrieval

systems into a unified scoring list. RRF computes each document's composite score as the sum of reciprocal ranks across all participating retrieval channels:  $RRF(d) = \sum (1 / (k + r(d)))$ , where  $r(d)$  is the ordinal rank of document  $d$  in a specific system and  $k$  is a constant parameter (empirically optimized to 60). Because it operates entirely on ordinal rank positions rather than uncalibrated raw relevance scores, RRF requires no scale normalization, score calibration, or supervised training.

**Advantages & Disadvantages:** The overarching advantage of RRF is its mathematical simplicity, zero-shot robustness, and remarkable capacity to consistently outperform Condorcet Fuse and complex learning-to-rank models across diverse TREC test collections. It inherently prioritizes consensus candidates appearing across multiple diverse search pipelines (e.g., lexical BM25 and dense vector spaces). Conversely, its primary disadvantage is that it discards the actual magnitude of similarity scores, treating minor and overwhelming margins of relevance identically. Furthermore, it assumes equal weighting across underlying retrievers, which can sub-optimally degrade precision when one retrieval channel exhibits substantially higher noise than another.

### **Sifei at SemEval-2026 Task 8: Hybrid Retrieval and Query Rewriting for Multi-Turn RAG**

**Methodology:** Meng & Ilvovsky (2026) developed a high-performing, training-free hybrid retrieval framework for multi-turn conversational RAG in the SemEval-2026 MTRAGEval benchmark. Their architecture addresses the drift in user intent, context limits, and conversational noise inherent to extended dialogue. The methodology integrates automated conversational corpus cleaning (filtering near-duplicates, markup canonicalization, and fragment pruning), LLM-based query rewriting that combines the raw last turn with an explicit standalone question rewrite ( $q = \text{last\_turn} \parallel \text{rewrite}$ ), hybrid lexical and dense retrieval over the cleaned corpus with fallback recovery, and cross-encoder reranking over the top candidates.

**Advantages & Disadvantages:** The system's primary advantage is its proven real-world efficacy, securing 3rd place out of 38 teams in Task A with an  $nDCG@5$  of 0.5453 (significantly beating the 0.4795 baseline) without requiring parameter fine-tuning. Concatenating the raw turn with the rewritten query robustly preserves conversational pronouns while infusing explicit entity grounding. The trade-off is the latency and financial cost of deploying an LLM query-rewriting API before retrieval, which introduces sequential dependencies that can increase end-to-end response times in latency-critical tutoring applications.

### **Decoupling Semantics and Logic: A Training-Free Coarse-to-Fine Pipeline for Video RAG**

**Methodology:** This work investigates zero-hallucination temporal grounding and long-video comprehension by decoupling semantic retrieval from cognitive logical reasoning into a two-stage cascaded architecture. In the first stage, a high-recall semantic pre-fetching module executes dense retrieval using exclusively high-fidelity visual summaries and global descriptions, isolating noisy raw OCR and ASR streams to prevent vector space corruption. In the second stage, an Adaptive, Iterative, and Reasoning-based (A.I.R.) filtering agent ingests full multimodal context to enforce strict persona alignment and prune irrelevant candidates, followed by Prompt Sculpting that forces the generator to output strictly formatted JSON responses containing precise chunk-level citations.

**Advantages & Disadvantages:** Decoupling feature pre-fetching from logical validation drastically reduces hallucinated timestamp citations while providing verified start/end video boundaries for downstream media players. The modular, training-free design operates natively with off-the-shelf vision-language models. However, the iterative cognitive filtering agent introduces multiple LLM reasoning calls over long multimodal contexts, causing high inference latency and substantial token consumption that may limit real-time conversational streaming.

### **MemGPT: Towards LLMs as Operating Systems**

**Methodology:** Packer et al. (2023) introduce MemGPT, an architecture inspired by traditional operating system hierarchical memory management, designed to overcome LLM context window constraints. MemGPT establishes a tiered virtual memory architecture that divides information into main context (in-context working memory, ephemeral chat buffer) and external context (out-of-context storage, including archival vector memory and user profiles). The model autonomously manages its memory hierarchy via function calls, paging context in and out, evicting temporary conversational noise, and executing explicit writes to update durable learner mastery and knowledge states.

**Advantages & Disadvantages:** MemGPT's primary advantage is its ability to support perpetual, multi-session conversational continuity while enforcing strict operational boundaries between ephemeral conversational turns and durable profiles. Its function-calling interface natively supports policy-driven updates, corrections, and explicit data deletions (supporting GDPR compliance). However, the system's reliability is bound to LLM reasoning fidelity; if an LLM fails to recognize when to trigger memory-management function calls, critical pedagogical updates can be missed or inappropriate context can be written to durable storage.

### **Trust or Abstain? A Self-Aware RAG Approach**

**Methodology:** Zhu et al. (2026) investigate the critical challenge of knowledge conflicts occurring when retrieved external context (CK) and internal parametric knowledge (PK) disagree or prove unreliable. They propose SABER (Self-Aware Belief Estimator for RAG), a training-free framework that instills self-awareness into frozen LLMs. SABER leverages multi-trace inference to extract conditional reasoning representations across both PK-only and CK-conditioned execution paths. It then uses two lightweight calibration predictors to construct a four-way decision space: trust parametric knowledge, trust contextual knowledge, trust either, or abstain.

**Advantages & Disadvantages:** SABER significantly elevates end-to-end generation accuracy on conflict-heavy datasets, and its risk-coverage curve Pareto-dominates prompt-based abstention baselines by offering a mathematically tunable trade-off between answering coverage and hallucination risk. The main limitation is the computational cost of multi-trace inference, which requires sampling multiple candidate reasoning paths per query, increasing inference token overhead.

## **Selective 'Selective Prediction': Reducing Unnecessary Abstention in Vision-Language Reasoning**

Methodology: Srinivasan et al. (2024) address over-cautious abstention in vision-language models (VLMs) deployed in zero-tolerance environments. Standard selective prediction models often discard large volumes of correct predictions when confidence scores dip below arbitrary thresholds. The authors introduce ReCoVERR, an inference-time algorithm that actively queries the visual scene upon low-confidence predictions. ReCoVERR directs an LLM to pose targeted questions to the VLM regarding specific visual cues in the image, collecting high-confidence corroborating evidence to confirm or overturn the tentative answer before abstaining.

Advantages & Disadvantages: ReCoVERR successfully increases answering coverage by up to 20% across VQAv2 and A-OKVQA benchmarks across diverse models (BLIP2, InstructBLIP, LLaVA-1.5) without degrading system precision or increasing hallucination rates. The drawback is the added latency of recursive, multi-turn visual questioning loops, which adds computational overhead when handling ambiguous visual inputs.

## **RAGAS: Automated Evaluation of Retrieval Augmented Generation**

Methodology: Es et al. (2023) developed Ragas, an automated, reference-free evaluation framework designed specifically to benchmark RAG pipelines without requiring human ground-truth annotations. Ragas decouples evaluation into orthogonal dimensions: Faithfulness (evaluating whether generated statements are grounded in the retrieved context using LLM-based claim decomposition), Answer Relevance (measuring how directly the response addresses the prompt using reverse question generation), and Context Relevance (evaluating whether retrieved passages contain only necessary information).

Advantages & Disadvantages: The major advantage of Ragas is its fully automated, scalable nature, enabling continuous integration benchmarking during architectural iteration. Its component-level isolation allows engineers to distinguish retrieval failures from generator hallucinations. However, its primary disadvantage is its dependency on an underlying evaluation LLM (LLM-as-a-judge), which can introduce evaluator biases, self-preference tendencies, and sensitivity to prompt formulation. Furthermore, it lacks native support for temporal timestamp bounds or multimodal visual grounding evaluation.

## **Certiably Robust RAG against Retrieval Corruption**

Methodology: Xiang et al. (2024) introduce RobustRAG, the first defense framework offering certifiable mathematical robustness against retrieval corruption attacks (such as malicious passage poisoning). RobustRAG implements an isolate-then-aggregate defense: retrieved documents are partitioned into disjoint groups, independent candidate responses are generated from each isolated cluster, and an aggregation algorithm (using keyword voting or decoding consensus) merges candidate answers into a hardened output. This guarantees that an attacker controlling fewer than a bounded number of injected passages cannot arbitrarily dictate the generated response.

Advantages & Disadvantages: RobustRAG provides formal theoretical security guarantees against adaptive adversaries with full knowledge of the defense mechanism, setting a benchmark for secure RAG. The trade-off is the linear multiplier in inference cost, as the generator must execute multiple independent

forward passes over partitioned context groups, introducing latency constraints in real-time interactive tutors.

### **Trustworthy RAG: An Evaluation Agent for Detecting Misinformation and Knowledge Poisoning**

Methodology: Giri et al. (2026) examine the Security-Reliability Gap in RAG systems, where high semantic relevance scores fail to prevent the injection of poisoned misinformation. The authors introduce an Evaluation Agent middleware that executes before response generation. The agent integrates Natural Language Inference (NLI) factual verification, a five-signal poison detector with relevance-weighted aggregation, and a composite Trust Index:  $T = 0.4F + 0.35C + 0.25(1 - P)$ , augmented with a non-linear dampening function for high-contamination contexts.

Advantages & Disadvantages: Tested on TruthfulQA with Llama 3.3 70B, the Evaluation Agent achieves 91% accuracy and 100% precision, exhibiting 100% recall against instruction injection attacks and an ROC-AUC of 0.73–0.81. It blocks malicious advice before generation occurs. Its limitation is vulnerability to subtle in-place entity swaps and semantic weakening, which evade NLI classification, as well as the need for per-LLM threshold calibration to balance clean accuracy with threat detection.

### **Hidden-in-Plain-Text: A Benchmark for Social-Web Indirect Prompt Injection in RAG**

Methodology: Guo & Wei (2026) present OpenRAG-Soc, an empirical benchmark and testing harness designed to evaluate RAG vulnerability to indirect prompt injection attacks embedded within user-generated web content. The benchmark models attacks that survive ingestion pipelines (e.g., hidden markdown payloads, CSS concealment, zero-width Unicode injection). It measures rank shifts in sparse and dense retrievers, evaluates defense mitigations (HTML sanitization, Unicode normalization, attribution gating), and calculates attack success rates at generation time.

Advantages & Disadvantages: The benchmark standardizes realistic, reproducible red-teaming evaluations across ingestion and generation layers, providing actionable metrics for web-scale RAG hardening. The main limitation is that the benchmark focuses primarily on textual web formatting exploits; indirect injection vectors embedded within visual keyframe imagery, slide diagrams, or OCR bounding boxes remain unaddressed.

### **VideoRAG: Retrieval-Augmented Generation with Extreme Long-Context Videos**

Methodology: Ren et al. (2025) introduce VideoRAG, a purpose-built framework for processing and reasoning over extreme long-context multimedia repositories. VideoRAG employs a dual-channel architecture: (1) graph-based textual knowledge grounding that extracts entities, captions, and ASR transcripts to map cross-video semantic dependencies into an evolving knowledge graph, and (2) multimodal context encoding that maps visual frames into shared embedding spaces using encoders like CLIP or ImageBind. A multi-channel retrieval paradigm integrates graph-traversal search with visual vector retrieval, followed by LLM-based clip filtering to select salient video moments.

**Advantages & Disadvantages:** VideoRAG overcomes context window exhaustion over multi-hour video collections, demonstrating superior performance on the 134-hour LongerVideos benchmark across comprehensiveness, empowerment, and trustworthiness metrics. Its dual-channel design preserves visual context while maintaining semantic structure. The primary limitation is its high offline indexing complexity: constructing and maintaining the multi-modal knowledge graph across extensive lecture archives requires substantial compute, token costs, and storage overhead.

### **TV-RAG: A Temporal-Aware and Semantic Entropy-Weighted Framework for Long Video Understanding**

**Methodology:** Cao et al. (2025) present TV-RAG, a training-free framework designed to mitigate temporal blindness and semantic drift in Large Video Language Models (LVLMs). TV-RAG incorporates two core mechanisms: (1) a time-decay retrieval module that applies explicit temporal offset weighting to similarity computations, prioritizing multimedia contexts aligned with narrative progression, and (2) an entropy-weighted keyframe sampler that selects evenly spaced, information-dense frames by calculating visual and linguistic semantic entropy, pruning redundant static frames without human intervention.

**Advantages & Disadvantages:** TV-RAG is completely training-free and model-agnostic, functioning as a lightweight plug-and-play module that enhances open-source LVLM backbones across Video-MME, MLVU, and LongVideoBench. Its entropy-based frame selection dramatically reduces token overhead while improving answer accuracy. The disadvantage is that the temporal decay heuristic assumes linear topic progression; in educational lectures that frequently revisit earlier concepts, static temporal decay can penalize relevant historical slides.

# Corpus

Corpus for multi modal data:

- Selected publicly available Corpus: <https://ocw.mit.edu/>
- Selected Corpus license : <https://creativecommons.org/licenses/by-nc-sa/4.0/>

**About the MIT OCW Corpus:** MIT OpenCourseWare (MIT OCW) is an initiative by the Massachusetts Institute of Technology to publish all of the educational materials from its undergraduate and graduate-level courses online, making them freely accessible worldwide. It acts as an immense public repository of university-level content.

**How we can use this corpus for our project:** Because the project intersects heavily with evaluating models on long-video semantic comprehension and multi-modal contexts (as evidenced by tools like NotebookLM and frameworks like VideoRAG and TV-RAG), the MIT OCW corpus is highly valuable in several ways:

1. **Multi-Modal Richness:** MIT OCW provides synchronized data streams—full video lectures, closed-caption audio transcripts, on-screen slides, assignments, and reading materials. This allows us to train or evaluate models on cross-modal dependencies, ensuring that our AI models do not suffer from "temporal blindness" and can cross-reference what is spoken in the audio with what is visually written on a whiteboard or slide.
2. **Evaluating Video RAG and Temporal Navigation:** The extreme long-context nature of university lecture videos offers an ideal testbed for evaluating frameworks like *VideoRAG* and *TV-RAG*. It allows us to benchmark a model's ability to navigate sub-second video frames, filter redundant keyframes, and ground generative responses accurately inside lengthy educational recordings.
3. **Document Evolution and Version Control:** We can leverage the evolving structure of MIT OCW's syllabi, problem sets, and course materials across semesters to effectively test *VersionRAG*, evaluating the models on how well they adapt to multi-version comparative tracking and document temporal updates.

## Detailed analysis of the license:

- **Creative Commons Framework:** The MIT OpenCourseWare (OCW) dataset is distributed under the Creative Commons Attribution-NonCommercial-ShareAlike (CC BY-NC-SA 4.0) license, which aligns perfectly with the scope of this academic project without risk of violation.
- **Non-Commercial (NC) Compliance:** The multimodal AI tutor is being developed purely for academic research rather than monetary gain or enterprise deployment, strictly satisfying the non-commercial constraint.
- **Attribution (BY) Compliance:** The system's core architecture inherently fulfills attribution requirements, as the timestamp-grounding methodology automatically

generates explicit citations and time intervals for every piece of synthesized evidence, crediting the original source.

- **ShareAlike (SA) Compliance:** To comply with the ShareAlike clause, any derivative multimodal datasets, code, or system architectures released publicly as a result of this research will be distributed under the identical CC BY-NC-SA 4.0 license, ensuring full legal and ethical compliance.

Github Repository: <https://github.com/SGowthamReddy2/Multimodal-evidence-grounded-AI-tutor>

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